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SOC

Analyst L1 For Free

Defronix Cyber Security → <https://defronix.com>

WINDOWS ARCHITECTURE

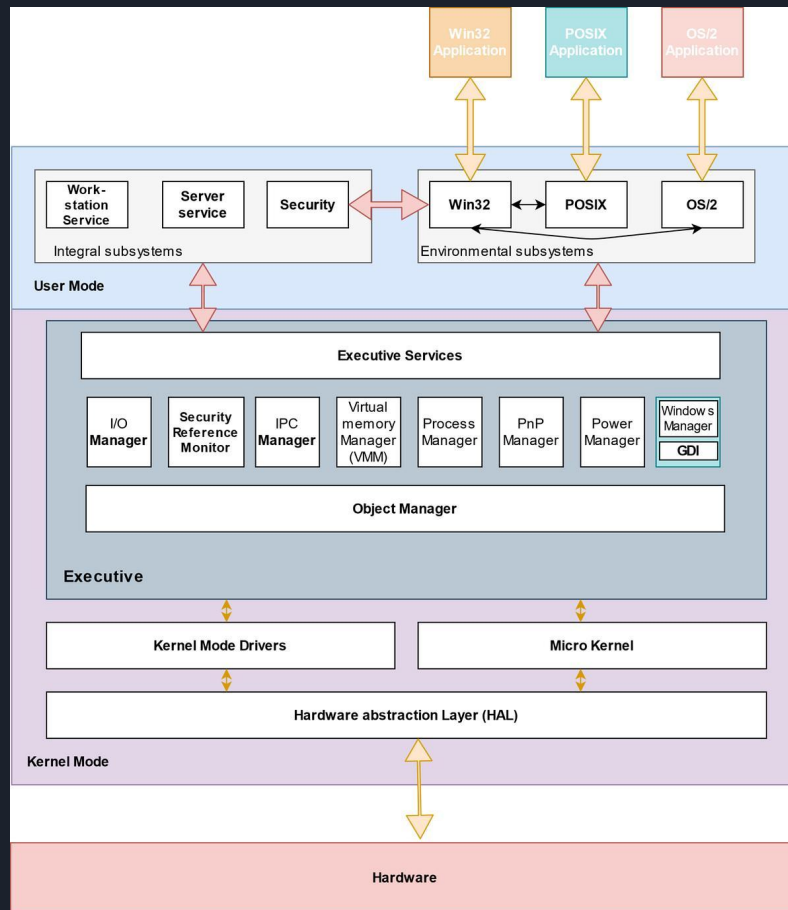


What is Windows Architecture?

Windows architecture defines:

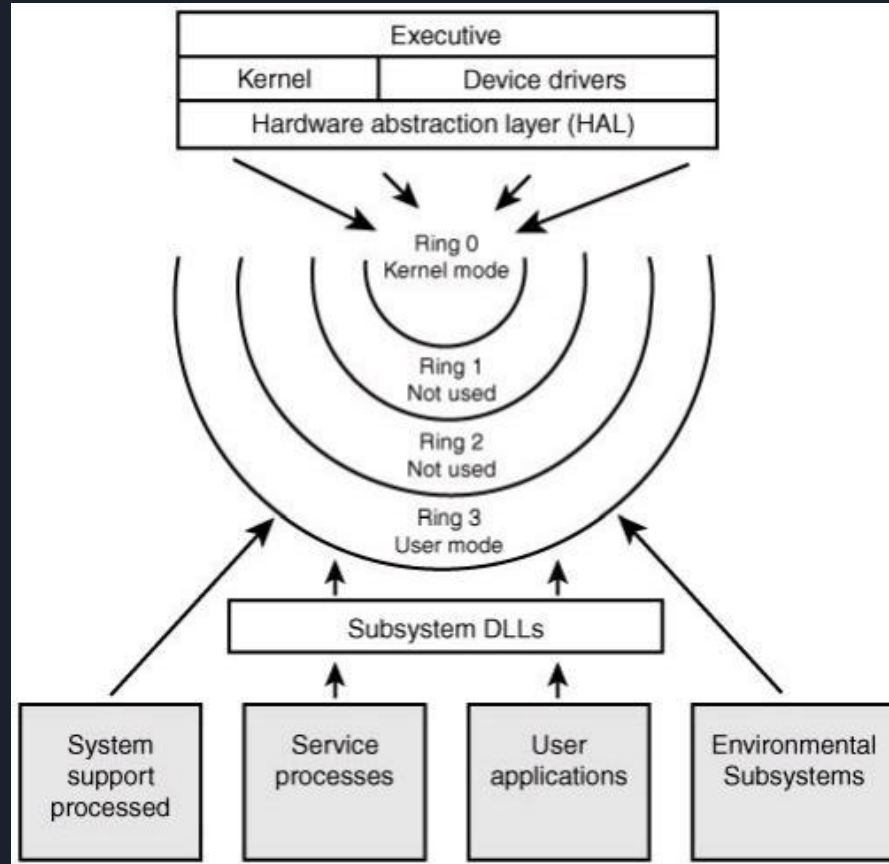
- How processes run
- How memory is handled
- How users authenticate
- How security is enforced

In SOC, this = **where attacks happen**



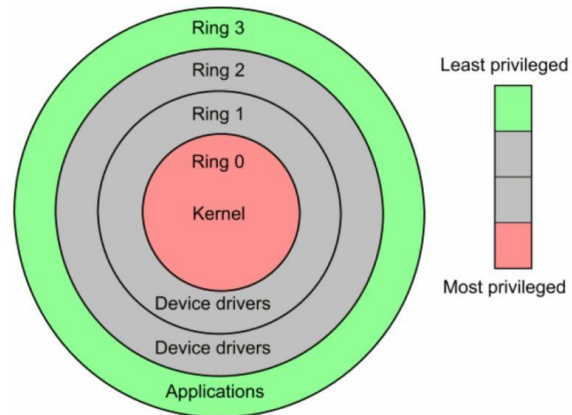
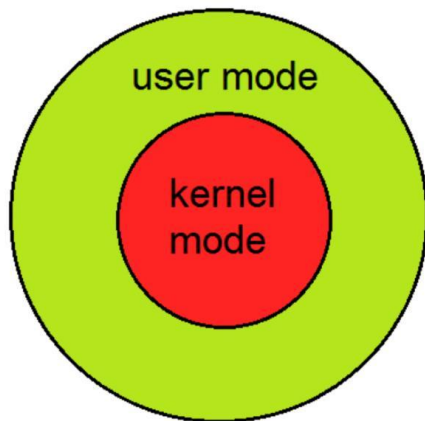


WINDOWS ARCHITECTURE





User Mode and Kernel Mode: Understanding Operating System Fundamentals



WINDOWS ARCHITECTURE

(Two Core Modes)



User Mode

Runs:

- Applications (Chrome, Word, etc.)
- User processes
- Some services

Restrictions:

- Cannot directly access hardware
- Cannot modify kernel

SOC Relevance:

- Most malware **starts here**
- Phishing → Word → PowerShell → Execution

Kernel Mode

Runs:

- Windows Kernel
- Drivers
- Memory manager

Power:

- Full system access

SOC Relevance:

- Rootkits operate here
- Kernel compromise = **complete system control**

WINDOWS ARCHITECTURE (Windows Core Components)



Windows Kernel

Handles:

- CPU scheduling
- Interrupts
- Thread execution

SOC Insight:

- Rarely directly attacked
- But if compromised → stealth malware

Executive Layer

Process Manager :

- Creates processes

👉 SOC:

- Detect abnormal process spawning

Memory Manager :

- Allocates memory

👉 SOC:

- Fileless malware lives here

WINDOWS ARCHITECTURE (Windows Core Components)



Executive Layer (cont..)

I/O Manager :

- Handles disk + network

👉 SOC:

- Detect abnormal file/network activity

Security Reference Monitor (SRM):

👉 Core security engine

Checks:

Can this user access this object?

SOC Use:

- Privilege escalation
- Unauthorized access

WINDOWS ARCHITECTURE (Windows Core Components)



HAL (Hardware Abstraction Layer)

- Interface between OS & hardware

👉 SOC:

- Used in advanced rootkits

Device Drivers

- Communicate with hardware

Threat:

- Vulnerable drivers → privilege escalation
- Signed driver abuse

WINDOWS PROCESS ARCHITECTURE



A Windows process is an executing instance of a program, acting as a container for resources like memory, handles, and threads needed to run applications.

It is the fundamental unit managed by the operating system, identified by a unique Process Identifier (PID)

**Process is a running program
with:**

- PID
- Memory
- Threads

WINDOWS PROCESS ARCHITECTURE

(MUST-KNOW SYSTEM PROCESSES)



smss.exe

- First user-mode process

wininit.exe

- Starts system

services.exe

- Manages services

lsass.exe 🔥

Handles:

- Authentication
- Credential storage

Attack:

- Mimikatz → dumps credentials

svchost.exe

- Hosts services

SOC:

- Malware hides here

SOC Detection

winword.exe → powershell.exe →
cmd.exe

Malicious macro execution



Virtual Memory

Each process gets isolated memory

SOC Threats

- Process injection
- DLL injection
- Fileless malware

WINDOWS AUTHENTICATION FLOW



User Login → LSASS → Authentication → Access Token

Access Token

Contains:

- User identity
- Privileges

SOC Use:

- Token theft → privilege escalation

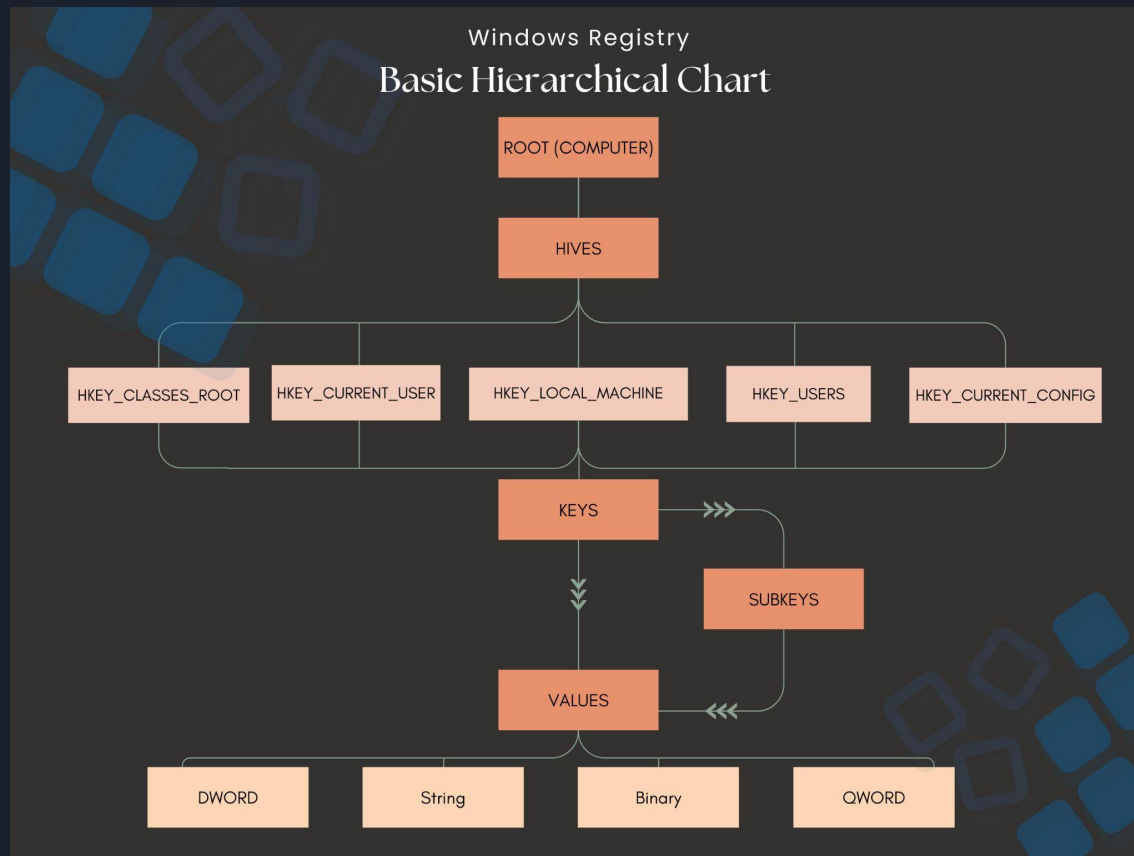
WINDOWS REGISTRY



What is Windows Registry?

Registry = Central database storing:

- System configuration
- User settings
- Application data



WINDOWS REGISTRY



Registry Editor

File Edit View Favorites Help

Computer\HKEY_CURRENT_USER\Microsoft\Windows\CurrentVersion\Explorer\StartupApproved\Run

Name	Type	Data
(Default)	REG_SZ	(value not set)
OneDriveSetup	REG_BINARY	04 00 00 00 00 00 00 00 00 00 00

Root Keys (red text, pointing to HKEY_CLASSES_ROOT and HKEY_CURRENT_USER)

Keys (orange text, pointing to the Microsoft key)

Subkeys (green text, pointing to the Windows, CurrentVersion, Explorer, StartupApproved, and Run keys)

Values (blue text, pointing to the (Default) and OneDriveSetup values)

WINDOWS REGISTRY



Registry Structure

Root Keys (Hive)

Hive	Meaning
HKLM	System-wide
HKCU	Current user
HKCR	File associations
HKU	All users
HKCC	Hardware config

Hive → Key → Subkey → Value



WINDOWS REGISTRY

HOW REGISTRY WORKS

When system boots:

1. Windows reads registry
2. Loads configs
3. Starts services
4. Applies user settings

When app runs:

App → reads registry → loads config

Types of Values :

REG_SZ → String
REG_DWORD → Numbers
REG_BINARY → Binary

WINDOWS REGISTRY



REGISTRY & ATTACKS

Persistence :

Location:

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Attack:

malware.exe added here

Runs at every login

RunOnce Key

- Executes once

Used by malware

Services Registry

HKLM\SYSTEM\CurrentControlSet\Services

Malware installs as service



WINDOWS REGISTRY

REGISTRY & ATTACKS

WMI Persistence

- Advanced persistence

Registry Hijacking

- Replace legit path

SOC DETECTION

What to Monitor

Suspicious Entries

powershell.exe in Run key

Unknown Executables

random.exe auto-start

Unusual Paths

AppData\Temp\

WINDOWS REGISTRY



REGISTRY FORENSICS

What can you extract?

- Installed software
- User activity
- Execution history

WINDOWS ARCHITECTURE + REGISTRY ATTACK FLOW

Phishing → Word → PowerShell



Payload executes



Registry Run key modified



Persistence achieved



LSASS attacked



Lateral movement

WINDOWS REGISTRY



REGISTRY For SOC

REAL SOC EXAMPLE

Event ID: 4657
Registry Key Modified

Key:
HKCU\...\Run

Value:
malware.exe

Result : Persistence detected

FINAL SOC MINDSET

When analyzing Windows system:

Ask:

1. Which process ran?
2. What did it modify?
3. Any registry persistence?
4. Any credential access?
5. Any lateral movement?